FLOUR IMPACT LAB

Problem Set: Potential Energy and Impact Prediction

Name:		Section:	
Date:		Score:	/ 25

This problem set uses the Flour Impact Lab simulator. In the simulator, an object (ping pong ball, potato, or canned good) is dropped from a chosen height onto a flour-covered surface. Use the data below to answer the questions. Assume air resistance is negligible and $g = 9.8 \text{ m/s}^2$ unless stated otherwise.

Reference Data from the Simulator

Object	Mass (kg)
Ping pong ball	0.0027
Potato	0.150
Canned good	0.400
Pick in Three Object	Exact Mass

Drop Height Options
0.20 m
0.40 m
0.60 m

Landing surface for all trials: 2-inch layer of flour (acts as a soft, decelerating surface upon impact).

Formula: $PE = mgh$

Problem 1 — Potential Energy Calculation

A 2 kg object is selected in the simulator and dropped from a height of 0.60 m.

1. Calculate the gravitational potential energy of the object before it is dropped.

Problem 2 — Changing the Mass

Trial 1 uses a 1 kg object dropped from 0.60 m.
Trial 2 uses a 3 kg object dropped from 0.60 m.

2. Calculate the PE of both trials.
3. Which trial has greater potential energy? Explain why.

Problem 3 — Changing the Height

An object with a mass of 2 kg is tested at different heights.

Trial A: 0.20 m
Trial B: 0.60 m

4. Calculate the PE for both heights.
5. How does increasing height affect the energy before impact?

Problem 4 — Comparing Two Simulator Trials

Trial A:

Object = Canned good
Mass = 0.143 kg
Height = 0.20 m

Trial B:

Object = Ping pong ball
Mass = 0.0027 kg
Height = 0.60 m

6. Calculate the PE of each trial.
7. Which trial will likely create a larger impact on the flour?

Problem 5 — Designing a Trial (Synthesis)

Using the simulator's available choices, design a single trial (pick one object, one height) that you predict will produce the largest impact force on the flour.

1. State your chosen object, mass and height.
2. Calculate the predicted PE.
3. In 2–3 sentences, justify why this combination of object, mass and height should give the largest impact force, referring to the mass, object and height involved.
